

EXPERIMENT - 3

Dimensional Modeling and Star Schema Design using Microsoft Power BI

Report



Subject Name: Business Intelligence & Tools
Subject Code: 24CSH-341

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Aim

To design and implement a dimensional data model using Microsoft Power BI by transforming transactional data from the Global Superstore dataset into a Star Schema consisting of Fact and Dimension tables for efficient reporting and analytical processing.

Tools / Software Required

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- DAX (Data Analysis Expressions)

Theory

Dimensional Modeling is a data warehouse design technique used in Business Intelligence systems to support analytical processing. It organizes data into Fact Tables and Dimension Tables.

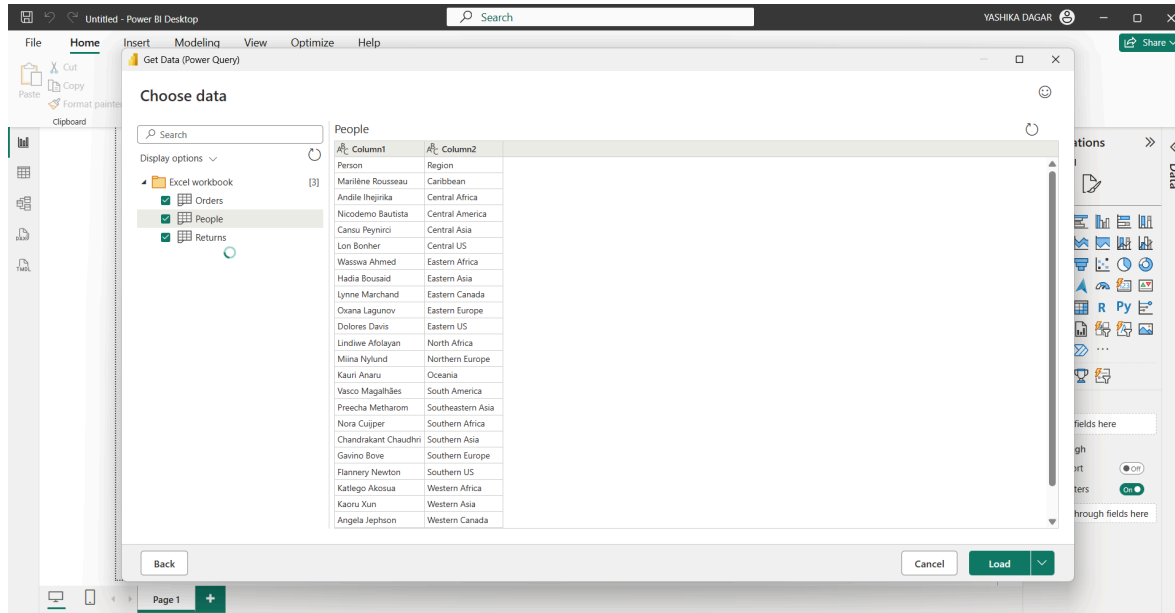
Fact Tables contain measurable business events such as Sales, Profit, Quantity, and Discount. Dimension Tables contain descriptive attributes such as Customer, Product, Region, and Date.

A Star Schema consists of a central Fact Table connected to multiple Dimension Tables. It improves query performance, simplifies reporting, and enhances scalability in BI environments.

Procedure

1. Load Dataset

- Opened Power BI Desktop.
- Imported Global Superstore Dataset.xlsx.
- Selected Orders, Returns, and People worksheets.
- Loaded and verified the imported tables.



2. Identify Fact and Dimension Tables

- Analyzed Orders for Sales, Profit, Quantity, and Discount.
- Identified Customer, Product, Region, and Date information.
- Categorized attributes into Fact and Dimension tables.

3. Create Dimension Tables

- Created Dim_Customer.
- Created Dim_Product.
- Created Dim_Region.
- Created Dim_Date.
- Removed duplicate records from dimensions.

4. Create Fact Table

- Created Fact_Sales with the required fields.
- Verified key columns required for relationships.

Table.TransformColumns(#"Removed Blank Rows",{{"Customer ID", Text.Proper, type text}, {"Customer Name",

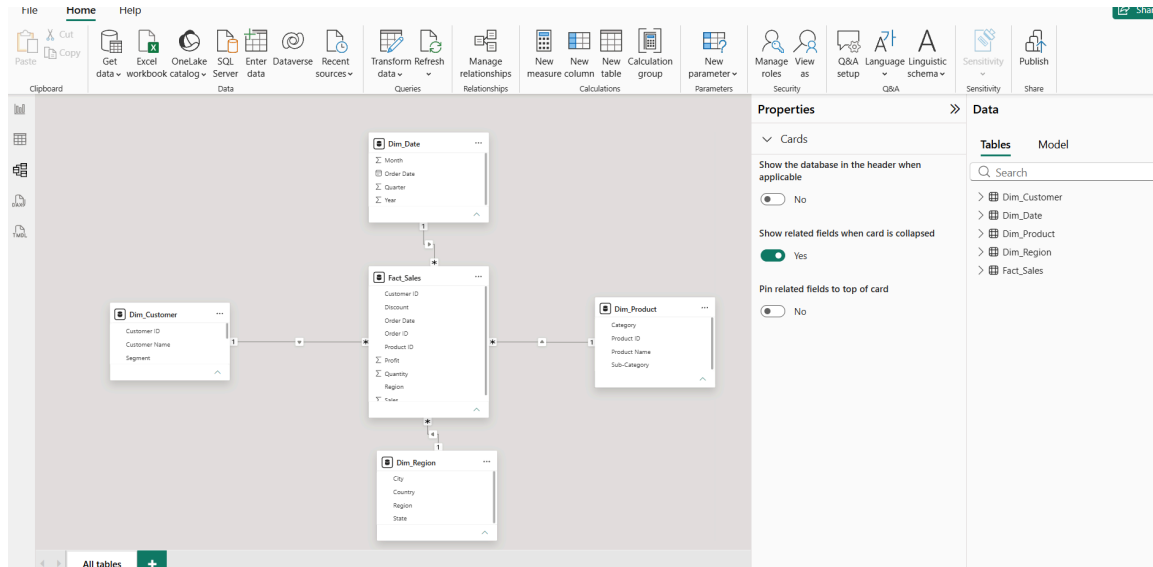
Customer ID	Customer Name	Segment
Ca-120551	Cathy Armstrong	Home Office
Bd-116051	Brian Dahlen	Consumer
Aj-107801	Anthony Jacobs	Corporate
Gm-144551	Gary Mitchum	Home Office
Vb-217451	Victoria Brennan	Corporate
Lo-171701	Lori Olson	Corporate
Ss-201401	Saphira Shifley	Corporate
Aa-103751	Allen Arnold	Consumer
Bg-110351	Barry Gonzalez	Consumer
Ah-105851	Angele Hood	Consumer
Cs-118451	Cari Sayre	Corporate
Dw-131951	David Wiener	Corporate
Rs-194201	Ricardo Sperren	Corporate
Ts-213401	Toby Swindell	Consumer
Jg-151151	Jack Garza	Consumer

5. Build Star Schema

- Opened Model View.
- Created relationships between Fact_Sales and dimensions.
- Configured relationships as Many-to-One.
- Arranged the tables into a Star Schema.

Manage relationships

From: table (column)	Relationship	To: table (column)	Status
Fact_Sales (Customer ID)	*—1	Dim_Customer (Customer ID)	Active
Fact_Sales (Order Date)	*—1	Dim_Date (Order Date)	Active
Fact_Sales (Product ID)	*—1	Dim_Product (Product ID)	Active
Fact_Sales (Region)	*—1	Dim_Region (Region)	Active



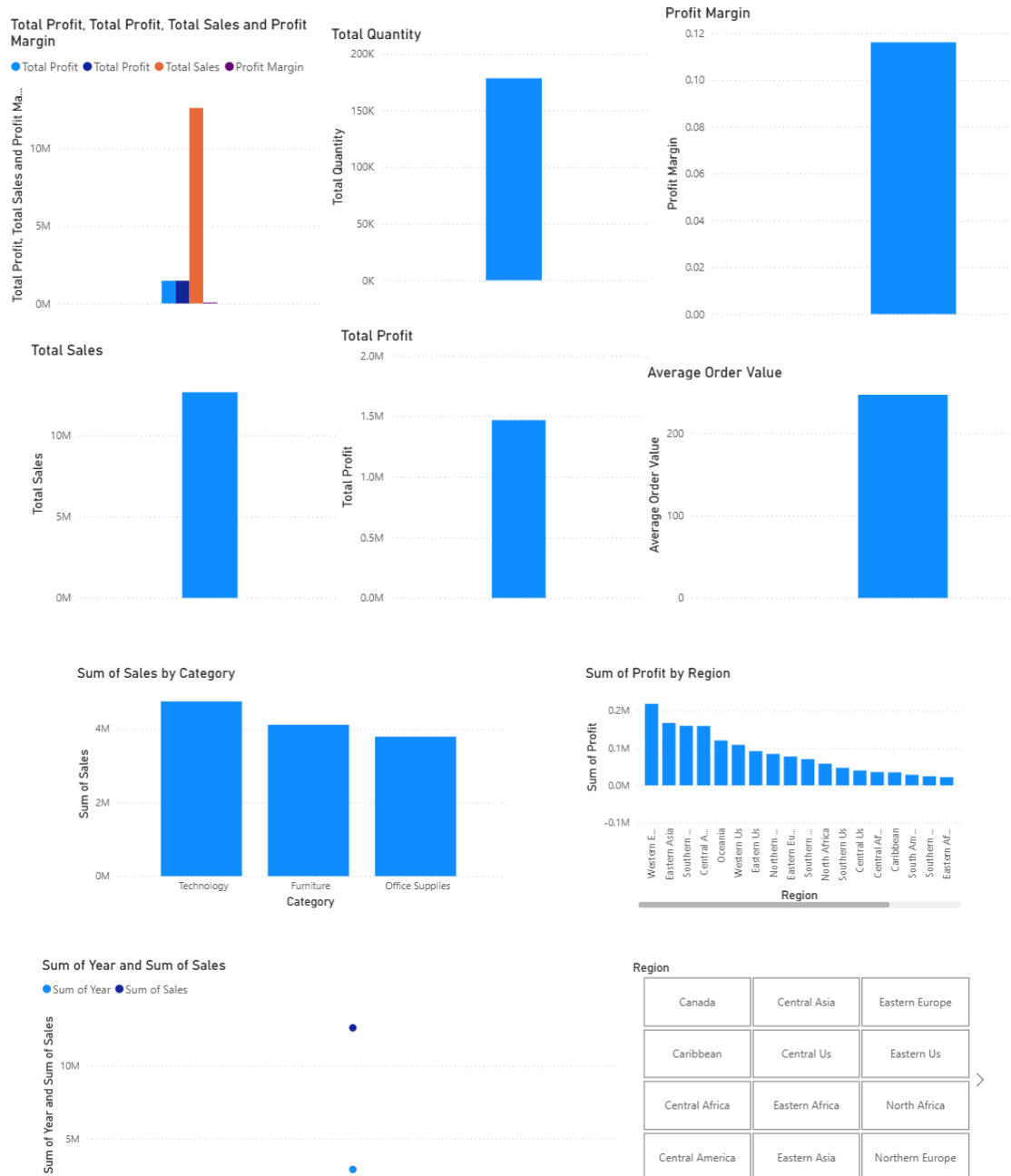
6. Validate Model

- Created Sales by Category, Profit by Region, and Sales Trend by Year.
- Created Total Sales, Total Profit, Profit Margin, and Average Order Value measures.
- Verified that dimensions correctly filter fact data.



Result

A dimensional model was successfully designed and implemented using Microsoft Power BI. A Star Schema was created consisting of one Fact Table and multiple Dimension Tables. Relationships and DAX measures were successfully implemented and validated using reports.



Learning Outcomes

- I understood dimensional modeling concepts and Star Schema architecture.
- I differentiated between Fact and Dimension tables.
- I designed a dimensional model using a real-world business dataset.
- I created and validated a Star Schema in Power BI.
- I developed DAX measures for analytical reporting.

Conclusion

This experiment demonstrated the implementation of dimensional modeling in Microsoft Power BI. The Global Superstore data was organized into Fact and Dimension tables and connected through a Star Schema. Visualizations and DAX measures were used to validate the model for analytical reporting.